

PAINTING DEFECTS CAUSES AND REMEDIES

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POPPING : It is closely scattered small blistering caused by solvent evaporated during curing.



CAUSES	REMEDIES
• Solvent evaporation is too quick from wet film	• Temperature and humidity should be maintained
• Oven Temperature is too high	• Give the right temperature
• Excessive paint coating between coats	• Paint coating should be as per DFT
• Air pressure is too low	• Give the right air pressure
• Flash off time is less	• Flash off time should be correct
• Improper paint viscosity	• Paint viscosity as per standard
• Gun distance is too close from surface	• Gun distance from surface should be 6-7inches

CISSING : Small round depressions in the surface of the paint



CAUSES		REMEDIES	
• Improper wiping of surface		• Properly clean the surface before painting	
• Improper mixing of paint		• Mixing of paint should be proper as per standard	
• Inadequate air pressure		• Give the proper air pressure	
• Oil or water from the compressed air		• Release the air from compressor and retake fresh air	

LOSS OF GLOSS (LESS GLOSS) : Dull or non glossy paint surface



CAUSES		REMEDIES	
• Hardner quantity is less used in paint mixing		• Hardner quantity should be proper in paint	
• Insufficient curing of primer before base coat		• Give sufficient time for curing	
• Less coating applied on the surface		• Coating should applied as per standard	

DUST : Particles projecting through the paint surface. It is usually small , irregular lifting the paint film caused by foreign matters (dirt) which can occur in different sizes , shapes, types and patterns.



CAUSES	REMEDIES
• Environmental dust	• Avoid the movement and close all doors
• IPA not proper wiping	• Proper IPA wiping as per WI
• Air pressure of spray gun is too low or high	• Set air pressure of spray gun as per standard
• Dirty paint booth	• Paint booth should be cleaned
• Clogged/Choked air filter inside booth	• Air filters needs to clean weekly
• Improper Booth balance	• Maintain the booth balance
• Improper filtration of paint and not changing the filter mesh on regular basis	• Proper filtration of paint and change the filter mesh every hour
• Jig dust due to air coming from spray gun	• Jigs should be stripped weekly
• Water flow on the screen is improper	• Proper water flowing on the screen
• Grease not applied on conveyor track	• Apply the grease on conveyor track
• Unsuitable work cloth	• Use only lint free cloth

LINT : Short threads that projected above the surface of paint film.



CAUSES		REMEDIES	
• Improper way of IPA wiping		• Proper IPA wiping as per WI	
• Lint free gloves is not using either while loading and IPA wiping		• Use only lint free gloves	
• Paint filter mesh tread is not removed properly		• Remove the thread of filter mesh	
• Faulty or choked air filter		• Filter should be cleaned	
• Improper IPA Cloth and not changing the cloth after 4 hours		• Use lint free IPA cloth and change the cloth in every 4 hours.	
• Improper booth balance		• Maintain booth balance	

ORANGE PEEL : The paint has an uneven texture which is similar to the skin of an orange.



CAUSES	REMEDIES
• Spray gun pressure is too low	• Set air pressure as per standard
• Spray gun is held too far or too close from the surface being sprayed	• Spray gun should be held at a particular distance from the surface being sprayed
• The paint applied is too lightly	• Paint apply as per standard
• Viscosity of paint is too low	• Maintain the viscosity
• Flash off time is too long between coats	• Give the proper flash off time between coats
• Solvent (hardner or thinner) is evaporated quickly	• Paint kitchen temperature should be maintained
• Booth temperature is too high	• Booth temperature should be maintained
• Spray gun pattern is too small	• Set the pattern after visual inspection
• Improper operator technique	• Proper way of painting

OIL MARK : Oil left out on the surface before painting



CAUSES		REMEDIES	
• Improper IPA wiping		• Proper IPA wiping	
• Paint mixed without stirred		• Before paint mixing paint should be stirred thoroughly	
• Paint mixing pot is not cleaned		• Always clean the jar in which paint mixed	
• Spray gun is not cleaned		• Clean the spray gun weekly	
• Moisture coming from air compressor		• Release the air inside the air compressor	

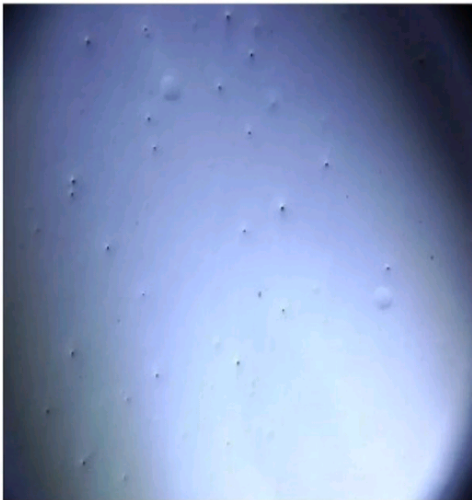
PAINT SAGGING/RUN DOWN

: Flowing of paint downward as soon as paint applied on the surface and forming a wavy appearance



CAUSES	REMEDIES
• Spray gun pattern is too close	• Set the pattern of gun after visual inspection
• Spray gun flow is excessive	• Set the flow of gun as per standard
• Improper paint mixing (too much thinner)	• Proper mixing of paint
• Sensor of Auto spray gun is not working continuously	• Sensor should be cleaned
• Air pressure of spray gun is too low	• Set the air pressure as per standard
• Improper jig rotation	• Jig rotation should be proper
• Improper paint viscosity	• Maintain the viscosity as per standard
• The flash off time between coats is not long enough	• Flash off time should be specified between the coats

PIN HOLE: Formation of tiny deep holes in wet paint film during painting and drying of the part



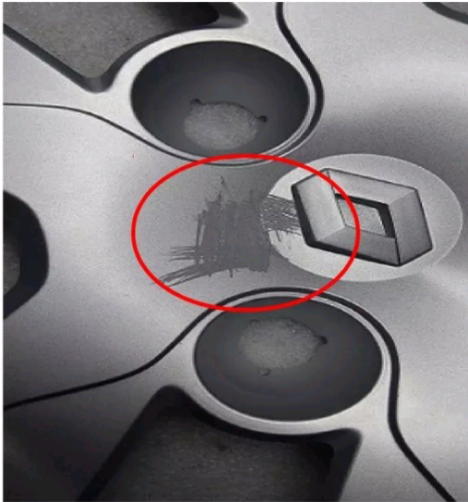
CAUSES	REMEDIES
• Improper drying time and oven temperature	• Give the proper drying time and Verify the oven temperature
• Improper paint mixing	• Mix the paint as per spec
• Improper flash off time	• Flash off time should be specified
• Improper IPA wiping	• Proper IPA wiping as per WI
• Poor operator technique (not wearing PPE)	• Operator should wear proper PPE
• Moisture coming in air	• Release the air from compressor
• Humidity is high inside the painting booth	• Humidity should be maintained within spec
• Temperature not maintained inside the painting booth	• Temperature should maintain within spec

SCRATCH : Scratch on the surface before or after the painting



CAUSES	REMEDIES
• Poor handling of parts	• Proper handling of parts
• Part kept on part without poly cover	• Keep the part in poly cover
• Wrongly emery applied	• Apply emery as per procedure defined
• Part removed from jig due to uneven jigs	• Identify and correct the uneven jigs
• On conveyor jig is touching during movements	• Avoid jig movements
• Part touching corner area of the trolley MS Clamps	• Foam apply on corner area of the trolley

PAINT PEEL OFF : Loss of adhesion between the coats. Inadequate bond between the individual coats.



CAUSES	REMEDIES
Improper IPA Cleaning	Proper IPA wiping
Improper paint mixing (hardner quantity is less)	Mix the Paint as per spec
Drying temperature is not maintained as per spec	Verify the drying temperature as per spec
Excessive thickness of paint coating	Apply paint as per standard coating
Improper flash off time	Give the proper flash off time
Spray gun nozzle is choked	Clean the gun nozzle every 3 hours
Primer missing because of sensor not working or either prime finished in paint container	Verify the working of sensor and availability of primer inside the container
Adhesion failure of primer due to too smoothness of surface	Apply emery on smooth surface before loading
Less DFT through coats	Maintain the DFT as per spec
Incorrect primer has been used	Use suitable adhesion primer

PAINT DROP/SPIT: When spray gun nozzle is choked some droplets of paint fall down on wet film surface



WATER MARK : Water running on the screen fall down on the part



CAUSES	REMEDIES
• Spray gun nozzle is choked	• Clean the gun nozzle every 3 hours
• Spray gun is not cleaned	• Clean the spray gun
• Air pressure is too low	• Set the air pressure as per spec

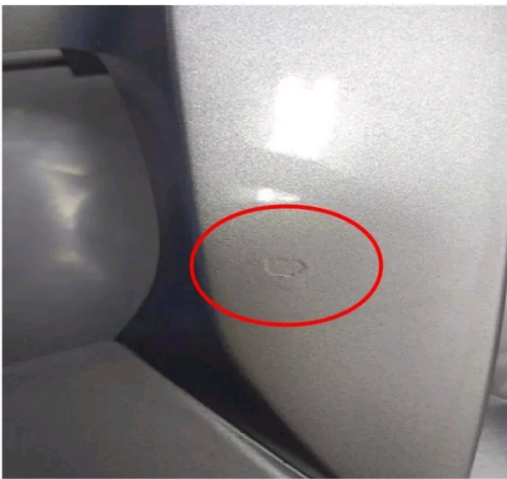
CAUSES	REMEDIES
• Improper water circulation on the screen	• Water circulation should be proper
• The surface of screen is not smooth	• Apply tak peelable /grease on the surface of screen
• Vibration inside the painting booth	• Vibration inside painting booth should be less
• Water circulating pump is not working properly	• Water circulating should be working properly

UNCOVER: Paint not applied (or missing) on some portion of the part



CAUSES	REMEDIES
• Uneven jig	• Jig should be perfect
• Jig rotation improper	• Jig rotation should be proper
• Sensor not working	• Verify the working of sensor
• Part loading on jigs is improper	• Part loading on jigs should be in proper way
• Booth balance improper	• Maintain the booth balance
• Paint applied is too light	• Apply the paint as per spec
• Poor gun setting	• Set the gun as per standard
• Air pressure of spray gun is too high	• Set the air pressure as per spec
• Poor operator (painter) technique	• Give the training to the painter for proper way of painting

DENT : It is a depression scratched into the surface made by pressure or blow

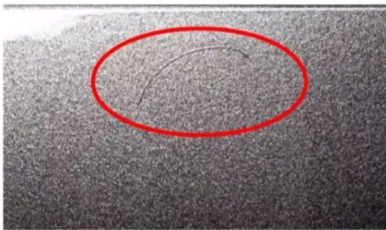


CAUSES	REMEDIES
Poor handling of the parts	Handling of parts should be done in proper way
On conveyor jig is touching during movement	Avoid such movements of jig
Part touching corner area of the trolley MS Clamps	Foam apply on corner area of the trolley
Part kept on part without using polycover	Use polycover to keeping the parts
Wrong method of emery applying on the surface of the part	Apply emery as per WI
Loading operator poor technique (sometimes dent available in molded part also)	Fully check the part before loading

EMERY MARK : Sanding scratches are visible on the painted surface



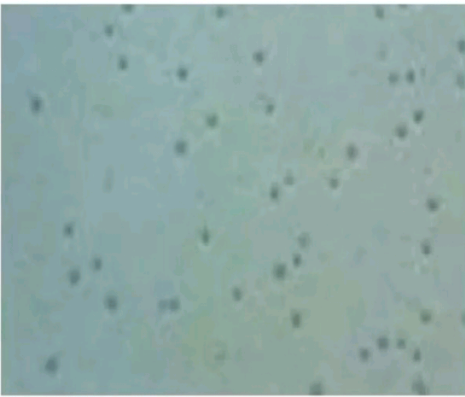
HAIR MARK : Hair fall down on the surface before drying of the part



CAUSES	REMEDIES
• Improper way of sanding	• Apply emery as per WI
• Too coarse sanding paper is used	• Use only recommended emery paper

CAUSES	REMEDIES
• Proper PPE's not wearing	• Wear PPE's before entering inside paint booth
• Movement of man inside booth is excessive	• Avoid movement inside booth

BLISTER : Lifting of the paint film from the underlying surface, which appears as bubble or blister in the paint



CAUSES		REMEDIES	
• Improper paint mixing		• Paint mixing should be as per spec	
• Improper drying time and oven temperature		• Give proper drying time and temperature	
• Humidity inside booth is too high		• Humidity should be maintained inside booth	
• Flash off time between coats is less		• Give the proper flash off time	
• Excessive moisture content in air of top coat		• Moisture content should not be available in air	
• Inadequate painting booth temperature		• Painting booth temperature should be maintained	
• Improper IPA wiping		• IPA wiping should be proper	
• Lacquer kept in container for more time		• Don't keep the lacquer for more time	